

RIMER: A Shoulder-Mounted Remote Dialogue Facilitation Robot for Situated Reminiscence Therapy

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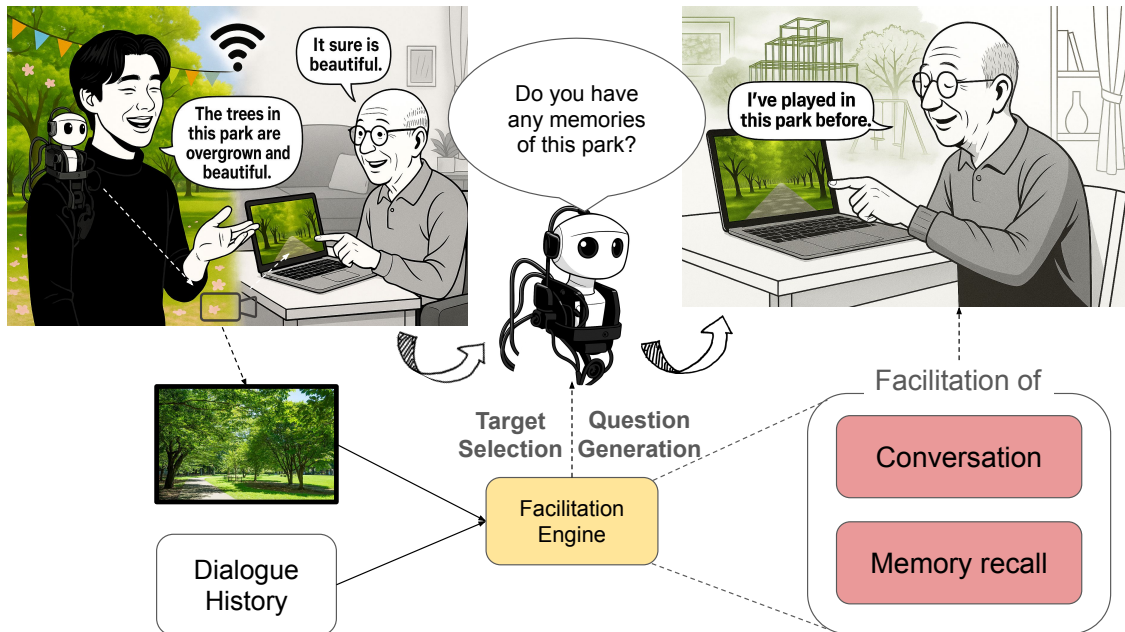


Figure 1: RIMER is a shoulder-mounted facilitation robot system designed to support remote reminiscence conversation by connecting an on-site local companion and a remote therapy recipient through live video streaming and AI-driven question generation. The local companion wears a camera-equipped robot that streams real-time visual information from the local environment, allowing the recipient to engage in conversation online while viewing the scene. RIMER leverages LLM to analyze both the conversation history and visual context, and it generates situated questions about current surroundings grounded in reminiscence therapy principles. Through the interaction loop, RIMER dynamically helps active conversation and supports cognitive and social engagement for therapy recipients.

Abstract

This paper proposes RIMER, a dialogue facilitation system specifically designed to support situation-aware remote reminiscence therapy using a shoulder-mounted robot. Reminiscence therapy is a psychological intervention that promotes cognitive improvement by encouraging individuals to recall and talk about their past experiences. Traditional reminiscence therapy typically involves face-to-face conversations guided by pre-selected photographs. In contrast, RIMER dynamically captures a local companion's current



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surroundings through a camera mounted on the robot and generates situated questions based on the observed scene and dialogue history. This allows a remote therapy recipient to naturally recall memories associated with what they are presently seeing and recent conversations, enabling more spontaneous and contextually relevant reminiscence. The situated therapy is conducted remotely, without the need for physical co-presence. To evaluate the effectiveness of RIMER, we conducted a comparative study with three groups: one using RIMER with reminiscence support, one with facilitation but without reminiscence, and one with no facilitation. We measured dialogue volume during the sessions and analyzed the effects of each condition on facilitation and memory recall. Results showed that the group using RIMER exhibited more conversation and stronger memory recall than the other groups.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; • **Computing methodologies** → **Artificial intelligence**; • **Hardware** → **Robotics**.

Keywords

Situated reminiscence therapy, Shoulder-mounted robot, Remote facilitation, Large language model, Human-robot interaction, Memory recall support, Socially assistive robotics, Dementia prevention, Conversation systems, Human-centered AI

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1 Introduction

In today's aging society, dementia has become a serious global issue. As of 2019, approximately 55.2 million people worldwide were estimated to be living with dementia, with the associated social cost amounting to 1.3 trillion USD annually [25]. Around 50% of this cost stems from unpaid informal care provided by family members and others, posing significant burdens on healthcare and welfare systems. Furthermore, dementia is one of the leading causes of years lived with disability (YLD) among those aged 60 and older and is a major contributor to mortality and disability worldwide. To address these challenges, international efforts have been underway to promote prevention, intervention, and care for dementia [17].

Reminiscence therapy, originally proposed by Butler [6], is a psychological approach aimed at enhancing psychological and cognitive functioning through the process of recalling past memories. It has demonstrated positive effects for people with dementia, helping to improve self-esteem, reduce feelings of loneliness, maintain or even enhance cognitive functioning [26]. Facilitating conversations rooted in personal memories has proven especially effective.

Traditionally, reminiscence therapy has been conducted using analog methods. Visual and auditory stimuli including old photographs, music, and personal belongings, are used to evoke memories. These memories are then verbalized and shared in face-to-face

interactions. However, more recently, digital technologies such as ICT, robotics, and virtual reality (VR) have enabled new forms of reminiscence support. For example, Moon et al. [19] reported a digital reminiscence therapy app reduced depressive symptoms and enhanced engagement. Yuan et al. [28] demonstrated the acceptability and efficacy of a robot with social dialogue functions for supporting memory recall in elderly individuals with dementia. Niki et al. [21] reported that VR-based reminiscence alleviated anxiety without inducing serious side effects, functioning as a safe intervention.

However, several limitations remain in existing digital reminiscence systems. Many web-based approaches continue to rely heavily on human facilitators, which not only imposes a significant burden but also introduces variability in support quality depending on the facilitator's experience and skills. Additionally, existing digital systems often employ fixed and one-way dialogue structures that hinder natural and flexible conversational development. Moreover, static stimuli such as pre-selected photos or cards also fail to convey a sense of immediacy in the conversation, limiting the situatedness in the interaction. If conversational systems could instead dynamically respond to the context and the surrounding environment, both user engagement and the effectiveness of reminiscence therapy are expected to be greatly enhanced.

To address these challenges, this paper proposes RIMER (Remote Interactive Memory Enhancement Robot), a shoulder-mounted facilitation robot system designed to support reminiscence therapy (Figure 1). RIMER is designed for remote reminiscence therapy involving a remote therapy recipient and an on-site local companion, who provides live situated media via first-person video. While streaming this media, RIMER continuously tracks the situational and conversational context and automatically generates situated questions that facilitate active, open-ended, flexible conversations and trigger spontaneous memory recall. With skill-independent facilitation powered by a large language model (LLM), RIMER mitigates the burden and variability of human facilitators. Leveraging live media to generate questions enables immediacy that was absent in the static, pre-selected stimuli employed by prior systems.

The rest of this paper is structured as follows. Section 2 reviews related work. Section 3 details the architecture and features of RIMER. Section 4 describes the experimental evaluation and discussion of RIMER's effectiveness. Section 5 concludes this paper.

2 Related Work

Reminiscence therapy is a psychosocial intervention that enhances cognitive and psychological well-being by encouraging individuals to share past experiences. To stimulate recall, visual and auditory cues such as old photographs, music, and familiar objects are often used [7]. Such interventions are typically delivered through individual or group conversations, and the use of media including paintings and videos has been shown to elicit deeper reminiscence. Theoretical frameworks such as Integrative and Instrumental Reminiscence provide structured approaches tailored to psychological needs, contributing to reductions in depressive symptoms and improvements in self-esteem [24]. Empirical research consistently

supports the effectiveness of reminiscence therapy: systematic reviews and meta-analyses confirm its impact on improving mood, cognition, and quality of life (QOL) [13, 23, 27].

With advances in digital technology, reminiscence therapy has expanded beyond its traditional face-to-face practice. Technology-supported interventions have gained traction in remote or resource-constrained contexts. Akhter et al. [3] introduced a Web-Based Reminiscence Therapy (WBRT) system using digital content and video calls, reporting benefits for managing behavioral and psychological symptoms of dementia (BPSD) and supporting consistent caregiving. Similarly, Moon et al. [19] demonstrated through an RCT that a digital reminiscence application not only alleviated depressive symptoms but also fostered greater participant engagement. Virtual reality (VR)-based approaches have also attracted significant interest. Mao et al. [18] conducted a systematic review showing gains in psychological well-being and cognitive function, though they noted limitations such as low interactivity. Other studies indicated that immersive VR reminiscence can reduce anxiety [21], and Ng et al. [20] confirmed its potential to enhance both emotional and cognitive outcomes.

Beyond reminiscence-specific systems, broader HCI and health-care technologies provide additional context. Abbasian et al. proposed *openCHA*, an open-source framework that integrates external data and multi-step reasoning to deliver personalized health queries [1]. Irfan et al. examined LLM-enabled companion robots for older adults, documenting challenges such as interruptions, repetitive or shallow responses, hallucinations, and outdated information—issues that raise concerns about trust and safety [11]. Telepresence robotics has also been explored as a strategy to mitigate loneliness. The DOMIROB project evaluated a 12-week in-home intervention, offering empirical findings and recommendations for deployment [12]. Research on wearable assistive devices emphasizes different design priorities: MIT’s E-BAR supports body weight and ambulation without requiring a harness [5], while Harvard’s soft robotic shoulder support assists patients with post-stroke or ALS-related impairments [4]. These advances illustrate progress in physical assistance, but they do not address conversational or memory-focused support.

Robotics has also opened new directions for reminiscence support. Yuan et al. [28] developed a robot-mediated therapy prototype for Alzheimer’s patients that elicited positive reactions such as smiling and singing. Liu et al. [16] combined robot-assisted rehabilitation with music therapy for stroke survivors, reporting improvements in self-efficacy and positive affect. Hsu et al. [10] showed that reminiscence interactions with robots can evoke a wide spectrum of emotions, underscoring the need for sensitive and empathetic design. Raingeard et al. [22] proposed musical robots as a non-pharmacological intervention to recommend personally meaningful music.

Despite these promising developments, several limitations remain. Many interventions still depend heavily on human facilitators [3], rely on static preselected content that weakens connections to users’ immediate experiences [19, 22], or suffer from limited conversational flexibility, often defaulting to repetitive dialogues [18, 28]. LLM-based systems face risks of shallow responses and outdated information [11]. Telepresence robots often target social connectedness but lack reminiscence-specific principles or

contextual visual input [12]. Wearable assistive devices, while advancing mobility and physical support, do not provide mechanisms for conversational or memory-oriented engagement [4, 5]. Together, these gaps highlight the need for systems that integrate contextual visual input, reminiscence-informed dialogue strategies, and adaptive interaction to better support memory recall and user engagement in remote contexts.

3 RIMER

3.1 Concept

This paper proposes RIMER, a shoulder-mounted robot designed to facilitate dialogue for situated reminiscence therapy. RIMER supports conversations between a remote therapy recipient and a local companion who walks on-site while wearing the robot, which captures and streams live, first-person visual input to the remote therapy recipient. It is specifically designed for a scenario where the companion walks around a place that the therapy recipient is familiar with. Notably, by referring to both the shared visual context and the ongoing conversational history, RIMER autonomously generates and delivers situated questions to stimulate memory recall. This allows the therapy recipient to naturally recall past experiences by simply answering questions that are grounded in the present scene.

3.2 Hardware

RIMER’s hardware is a shoulder-mounted robot worn by the local companion. The robot carries a forward-facing camera aligned with the wearer’s eye level, so the therapy recipient receives a live, first-person view that matches the local companion’s gaze [14]. While this video stream is relayed through the video-conferencing channel, the same camera also captures still images every five seconds; these images later serve as contextual input for question generation.

To broaden visual coverage and impart a lifelike presence, the robot’s head periodically rotates left and right. This motion lets it “look around,” gathering side views in addition to the forward view, and gives both users the sense that the system is actively surveying the surroundings. The combination of a shoulder-level vantage point, periodic head motion, and time-stamped image capture supplies rich, situated visual data to the facilitation engine described next, without imposing extra tasks on the local companion.

3.3 Facilitation Engine Architecture

The architecture of RIMER’s Facilitation Engine is illustrated in Figure 2. It consists of five core modules. The speech-to-text module takes the user’s spoken input and outputs transcribed text, which is stored as conversation logs. The local visual capture module outputs images captured by the camera mounted on the shoulder robot worn by the local companion. The question target selection module receives the conversation logs as input and determines the recipient of the next question. The question generation module takes as input the conversation logs, the captured images, and the selected question target, and generates contextually relevant questions. The question delivery module converts the generated questions into synthesized speech for output. To accommodate dynamic conversations and changes in the physical environment as

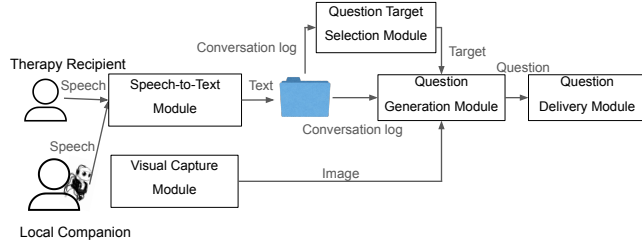


Figure 2: Architecture of Facilitation Engine

the local companion moves, each module operates independently and asynchronously.

Speech-to-Text and Visual Capture Module

The speech-to-text module transcribes spoken content from the two-person conversation using Google Speech-to-Text (GSTT). The local visual capture module uses the camera mounted on the shoulder robot worn by the local companion to share live video through a video conferencing tool. In addition to the video feed, still images of the local environment are captured at fixed time intervals. Since these images are used as contextual input for question generation, photos are taken every five seconds.

Question Target Selection Module

RIMER aims to maximize the memory recall of the therapy recipient while maintaining a natural and balanced conversational flow with the local companion. To achieve this, we designed a module that dynamically determines whom to address with each question. Figure 3 represents the logic of the module. As the primary target of reminiscence facilitation, the remote therapy recipient is initially asked more questions to encourage active recall. To prevent one-sided interactions, the local companion is also periodically engaged. The balance of this is controlled by constant parameters `NUM_ASK_RECIPIENT` and `NUM_ASK_BOTH`. Through our preliminary trial, we set them 3 and 5, respectively.

In addition, rather than relying solely on fixed thresholds for the number of questions, the module continuously evaluates the conversational context to decide whether further elaboration is necessary. Specifically, it assesses: (1) whether additional memory recall should be encouraged, and (2) whether the current topic warrants deeper discussion. If either condition is met, the system continues to prioritize the remote therapy recipient; otherwise, it may shift to involve both users or focus temporarily on the local companion. The assessment was implemented with an LLM.

Even though prioritizing the therapy recipient may seem sufficient from the standpoint of memory recall, the goal of facilitating natural conversation necessitates periodically engaging the local companion as well. Therefore, we define a maximum count `NUM_ASK_BOTH` to ensure that the local companion is asked at least once. When the combined count of questions directed solely to the therapy recipient and to both users reaches `NUM_ASK_BOTH`, the next question is directed exclusively to the local companion.

Question Generation Module

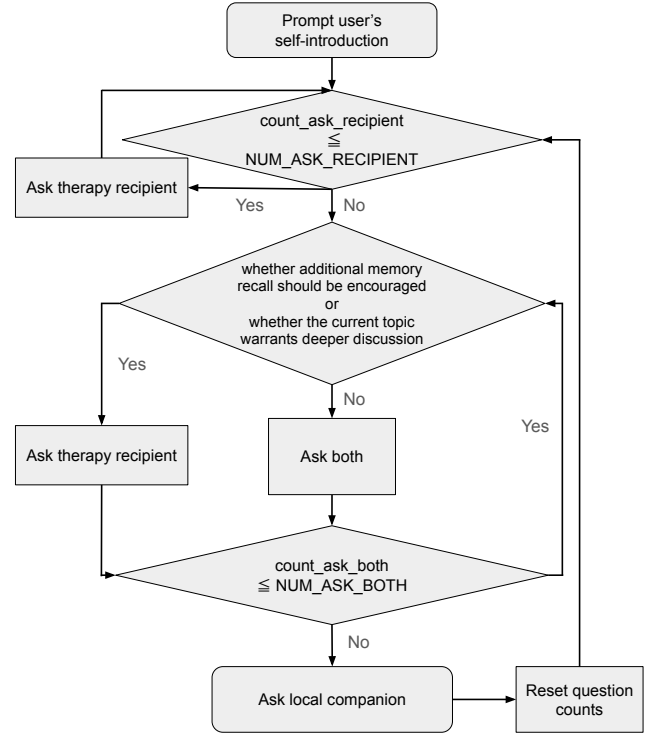


Figure 3: Flowchart for selecting the question recipient

The question generation module uses LLM to generate questions based on the conversation logs (from the speech-to-text module) and images (from the visual capture module). Prompts are tailored for each user: for the therapy recipient, the prompt instructs the model to generate reminiscence-based questions that facilitate memory recall; for the local companion, the prompt encourages the generation of conversational questions that support lively, flowing dialogue.

Figure 4 illustrates examples of generated questions under the theme of "school life," using GPT-4 [2] with and without reminiscence-based prompting. When reminiscence prompts are used, the resulting questions tend to be more concrete and specific, making it easier for the respondent to recall and share detailed experiences. For instance, a question like "What was your favorite school event?" not only refers to a personal memory but also evokes emotional associations. In contrast, questions generated without reminiscence guidance are more abstract, leading to general responses that may not promote extended conversation. For example, "What do you think about school lunches?" invites factual opinions rather than personal recollection. Thus, questions prompted by reminiscence guidance are expected to be more effective at encouraging episodic memory recall and deepening dialogue.

Question Delivery Module

The question delivery module converts the generated text questions into synthesized speech using Google Text-to-Speech (GTTS). If a period of silence between the two users exceeds a predefined threshold, the system interprets this as a conversational pause and

- Questions using reminiscence therapy**
- What was your most enjoyable school event during your student days? Who did you spend it with?
 - Could you share some memories with the first friend you made at school?
 - Was there a particular menu item in your school lunches or homemade bentos that you especially looked forward to?
- Questions without reminiscence therapy**
- Please tell me about school events.
 - How did you spend time with your friends at school?
 - What do you think about school lunches?

Figure 4: Examples of questions with and without reminiscence prompting under the theme of school life

outputs the synthesized question. To avoid misclassifying thoughtful silences as breakdowns, the threshold is set to four seconds. The synthesized voice is audible to both users.

4 Evaluation

4.1 Objectives

This experiment had two primary objectives. The first was to evaluate whether RIMER effectively facilitates two-person conversations across remote locations. The second was to assess whether RIMER enhances users' ability to recall personal memories.

4.2 Methodology

The experiment was conducted with nine university students (seven male, two female; aged 21–23 ($M = 21.7$, $SD = 0.71$)). To eliminate the influence of pre-existing relationships or prior knowledge between participants, which could affect conversational dynamics, all experimental pairs were composed of individuals meeting for the first time. As detailed later, the study was conducted under three different experimental conditions using a within-participants design. To maintain the “first-time meeting” constraint across all conditions, care was taken to ensure that no pair was repeated in any session.

While reminiscence therapy is typically applied to older adults, we first aimed to validate RIMER's mechanism for facilitating memory-related conversation in a controlled setting, using younger participants who could reliably engage with the questions and interaction flow. In addition, the prevalence of young-onset dementia (YOD) has been increasingly recognized. YOD affects 119 out of every 100,000 people aged 30 to 64 [9]. Furthermore, the diagnosis and support of YOD present unique challenges [15], and its social burden continues to grow [8]. Given these factors, we also considered the potential relevance of the situated reminiscence for younger populations.

To encourage memory recall in therapy recipients, the physical location used for the local companion's walk needed to be a place with personal significance to the therapy recipient. Accordingly, the experiment was conducted on the university campus attended by all participants. The local companion walked through this familiar campus while conversing with the therapy recipient. Moreover, it was assumed that the effects of facilitation would become more evident when conversations began to stagnate. Therefore, the walking route for local companions was designed to take approximately

20–30 minutes to complete, ensuring sufficient opportunity for natural pauses and system intervention.

At the beginning of the session, the robot initiated the interaction by saying, “I am a support system that facilitates conversation between two people. Please begin by introducing yourselves.” This prompted an initial round of introductions and helped establish a conversational flow. Afterward, the system allowed the conversation to continue naturally until it detected stagnation, at which point it presented a relevant question. At the end of the experiment, participants completed a questionnaire about their experience.

4.3 Conditions and Evaluation Metrics

This experiment was designed to evaluate the dual effects of RIMER: (1) its capacity to facilitate conversation and (2) its ability to stimulate memory recall through dialogue. These effects were assessed by establishing the following three experimental conditions.

Evaluating Memory Recall: To evaluate the effectiveness of RIMER's reminiscence-based facilitation on memory recall, we compared two conditions: one where facilitation was performed using prompts explicitly grounded in reminiscence therapy, and another where facilitation was applied without including any reminiscence-related content in the prompt for the question generation module.

Evaluating Facilitation Effectiveness: To assess whether conversation facilitation itself was effective, we compared conditions with and without facilitation. In the condition without facilitation, the system did not intervene at all during the interaction; instead, the local and therapy recipients freely engaged in conversation through the robot without any system-generated prompts.

Table 1 summarizes the experimental conditions. Note that this is not a full-factorial 2×2 design; rather, it is a partial design composed of three conditions. Specifically, the condition in which reminiscence is applied without facilitation is intentionally omitted, as the effect of reminiscence-based prompts does not appear without facilitation. Thus, this study focuses on comparing (Group 1) full reminiscence-based facilitation, (Group 2) non-reminiscence facilitation, and (Group 3) no facilitation, in order to evaluate both the impact of reminiscence and the effectiveness of facilitation.

Table 1: Experimental Conditions

	Reminiscence	Facilitation
Group 1	✓	✓
Group 2	×	✓
Group 3	×	×

Participants from Groups 1, 2, and 3 were asked to complete a post-experiment questionnaire. Responses were rated on a 7-point Likert scale ranging from 1 (“Strongly Disagree”) to 7 (“Strongly Agree”). The survey items are listed in Table 2. Questions Q1 to Q3 were designed to assess the impact of conversation facilitation (Conversational Activation), while Q4 focused on evaluating the promotion of memory recall (Memory Recall). In addition, participants were asked to explain their responses using open-ended questions listed in Table 3. Specifically, Q6 and Q7 explored reasons

behind participants' ratings of conversational activation, while Q8 investigated the context of memory recall. Because Group 3 did not use the facilitation system, questions that assumed system interaction (Q3, Q4, and Q8) were excluded for that group. To further assess the effect of facilitation by RIMER, we also recorded the number of conversational exchanges for each pair in every group. A single exchange was defined as one back-and-forth interaction between users, excluding minimal responses such as acknowledgments or interjections.

Table 2: Survey Items Using a 7-Point Likert Scale

Q1	Did you feel the conversation was engaging? (CA)
Q2	Do you think the system helped draw out conversation from both participants? (CA)
Q3	Was it easy to keep the conversation going? (CA)
Q4	Did you recall any past memories through the conversation? (MR)
Q5	Would you like to use this system again?

Note: CA = Conversational Activation, MR = Memory Recall.

Table 3: Open-Ended Questions for Reasoning

Q6	If you felt the conversation was engaging, why do you think that was the case?
Q7	If you felt that the system helped draw out conversation, how do you think it did so?
Q8	If you recalled any memories, what kind of memories did you recall?

4.4 Results

This section presents the experimental results for Groups 1, 2, and 3, based on the survey questions listed in Table 2 and the recorded number of conversational turns for each pair. Figure 5 illustrates the results of Questions 1 through 5 along with the number of conversational exchanges.

To evaluate the effectiveness of RIMER in facilitating conversation, we analyzed the responses to Questions 1, 2, and 3. Question 2 was not asked of Group 3, which did not use the system, so comparisons for this item are limited to Groups 1 and 2. To assess memory recall, we examined responses to Question 4. Since the degree of memory recall is influenced by whether the facilitation used reminiscence-based prompting, we compared Groups 1 and 2 for this item. Additionally, because memory recall was specifically targeted for therapy recipients, comparisons were made across remote therapy recipients in each group.

Q1: Did the conversation feel engaging?

A Kruskal–Wallis test revealed a statistically significant difference across the three groups ($p < 0.01$). Follow-up analysis using the

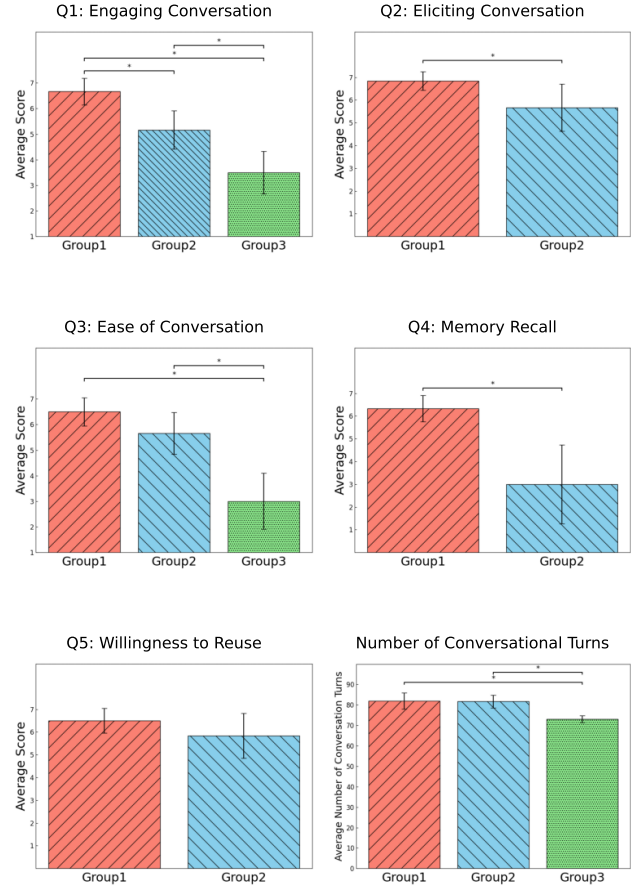


Figure 5: Survey responses to Questions 1-5 and the measured number of conversational exchanges.

Steel–Dwass test showed significant differences between all group pairs: Group 1 vs. Group 2, Group 1 vs. Group 3, and Group 2 vs. Group 3 ($p < 0.05$ for all).

Q2: Do you think the system helped draw out conversation from both participants?

A Wilcoxon rank-sum test was conducted between Groups 1 and 2. The test revealed a significant difference with $p = 0.0374 < 0.05$.

Q3: Was it easy to keep the conversation going?

A Kruskal–Wallis test again indicated a significant difference among the groups ($p < 0.01$). The Steel–Dwass post hoc test found no significant difference between Groups 1 and 2 ($p > 0.05$), but significant differences were observed between Group 1 and Group 3 and between Group 2 and Group 3 ($p < 0.05$).

Q4: Did you recall any past memories through the conversation?

A Wilcoxon rank-sum test was applied to the responses of therapy recipients in Groups 1 and 2. The test showed a significant difference with $p = 0.0495 < 0.05$. This suggests that reminiscence-based dialogue conducted remotely was effective and that RIMER successfully promoted memory recall among remote therapy recipients.

Q5: Would you like to use this system again?

A Wilcoxon rank-sum test between Groups 1 and 2 found no significant difference ($p = 0.230 > 0.05$).

Number of Conversational Turns:

A one-way analysis of variance (ANOVA) revealed a significant difference in the number of conversational exchanges among the three groups ($p < 0.02$). Tukey's HSD test showed no significant difference between Groups 1 and 2 ($p > 0.05$), but significant differences were observed between Group 1 and Group 3, and between Group 2 and Group 3 ($p < 0.05$).

4.5 Discussion

Discussion on Q1 (Conversational Activation)

Statistically significant differences were observed between Groups 1 and 3, as well as between Groups 2 and 3, indicating that the presence of conversational facilitation contributed to more active and engaging dialogue. The significant difference between Groups 1 and 2 further suggests that facilitation based on reminiscence therapy was especially effective in enhancing conversational liveliness. To better understand the factors underlying these results, we analyzed participants' open-ended responses to Q6, which asked for the reasons behind their ratings of conversational activation.

In Group 1, where questions were generated based on reminiscence therapy, participants frequently reported that the system supported the expansion of conversations centered around past memories. For example, one participant noted that when the topic of school trips arose, the system "asked questions aligned with the topic, such as about school trip memories." Others felt that "the system deepened the discussion based on each question" or that its topic suggestions "helped me think of things to talk about." Such comments indicate that the system was effective in prompting users to recall personal experiences connected to the current topic, which in turn facilitated the sharing of emotions and anecdotes from the past. In particular, for events like school trips, which both users had experienced and which evoked vivid memories, it became easier to have richer and more meaningful conversations.

In contrast, Group 2 received questions not based on reminiscence therapy, resulting in conversations that primarily focused on recent experiences. Most participants still felt that the facilitation "helped by introducing new topics when the conversation stalled," or that "when someone hadn't spoken much, the system directed a new topic to them." However, one participant noted they could talk extensively about the school area simply because they lived nearby, implying that the conversation tended to stay within the scope of familiar, recent topics. Another described the conversation as "casual and ongoing, but not particularly engaging." While facilitation enabled mutual sharing, the exchanges mostly stayed at the level of current updates and lacked the emotional depth of Group 1.

For Group 3, which received no facilitation, participants reported struggling to come up with topics themselves, with comments such as "There were no topics to talk about" and "Since no new topics were suggested, I had to come up with topics on my own." This lack of support led to increased cognitive load and fatigue. Nonetheless, some were able to engage in meaningful conversation when they discovered shared interests from past experiences, such

as American football or soccer—suggesting that facilitation is not strictly necessary if such common ground naturally emerges.

Taken together, these findings suggest that identifying and building upon shared experiences is critical to fostering engaging conversations. Especially when both participants have been deeply involved in similar activities—such as sports, as several comments indicated—dialogue tends to flourish. Thus, when a conversation continues for an extended period, it is likely that the topic resonates with the lived experiences of both users. Conversational facilitation that accounts for content can propose questions that further explore such shared topics, enhancing engagement beyond what is possible in unguided discussion. In particular, reminiscence-based facilitation, by highlighting past common experiences, can promote a stronger sense of connection and stimulate more lively, emotionally rich conversation.

Discussion on Q2 (Conversational Elicitation)

The statistically significant difference observed between Groups 1 and 2 indicates that reminiscence-based conversational facilitation was more effective in eliciting user dialogue than facilitation without such a basis. To explore the underlying reasons, we analyzed participants' responses to Q7, which asked why they felt the system helped elicit conversation.

Generating questions that align with the preceding dialogue appears to be key to effectively eliciting responses. Both groups made comments to this effect; a user in Group 1 noted that "the questions required thoughtful answers and were well aligned with the conversation context," while a user in Group 2 felt that "the questions felt so natural that I forgot which topics were system-generated." When system-generated prompts naturally continued the conversational trajectory, users reported being more likely to accept and respond to them smoothly. In this study, the system also incorporated environmental information from the local companion's surroundings. For example, one user in Group 1 commented that the system "brought up relevant topics from an unexpected angle," likely referring to prompts generated using visual input as well as conversation history. This integration of contextual data helped participants feel that the system was actively drawing out meaningful conversation. However, the significant difference between the groups suggests that other factors, beyond conversational alignment and environmental awareness, played a role.

The crucial difference lay in the content of the elicited topics. In Group 1, the system's questions helped retrieve deeply rooted personal memories. For instance, one user explained, "the system helped me recall student days, including amusing memories like typical high school experiences," and another stated, "when I struggled to express myself, the system helped me recall childhood play experiences." In contrast, Group 2's questions, while aligned with the dialogue, often focused on recent experiences. For example, one user felt that "the system asked deeper questions based on recent topics," and another described being asked "mostly about recent activities, like hobbies and places I visited." Furthermore, a participant in Group 2 noted that although "the system asked about everyday topics, it didn't expand the conversation beyond what we usually talk about." This highlights a general tendency to recall and discuss recent events more readily. As a result, users may have felt that the facilitation in Group 2 did not significantly expand the conversation beyond what they could have produced on their own.

These findings suggest that people naturally center conversations around recent experiences. However, by introducing reminiscence-based prompts, the system can steer the dialogue toward older, personally meaningful memories that might not otherwise surface in everyday interactions. This enables users to retrieve and articulate significant past experiences, potentially triggering a chain of associated recollections. Consequently, participants in Group 1—who received reminiscence-based prompts—were more likely to feel that new and meaningful topics had been drawn out of them, compared to those in Group 2.

Discussion on Q3 (Conversational Continuity)

The results of Q3 revealed significant differences between Groups 1 and 3, and between Groups 2 and 3. This suggests that the presence of conversational facilitation—regardless of whether it was based on reminiscence—helped participants maintain the conversation more easily. In contrast, there was no significant difference between Groups 1 and 2, indicating that the use of reminiscence specifically did not affect the perceived ease of continuing the dialogue. Similarly, the analysis of total utterance counts (Figure 5) showed significant differences between Groups 1 and 3, and between Groups 2 and 3, but not between Groups 1 and 2. These results further support the idea that conversational facilitation contributes to smoother progression of dialogue, while reminiscence content itself does not necessarily make conversations easier to sustain. One potential reason participants might find it difficult to continue a conversation is the frequency of conversational stalls. RIMER was designed to monitor periods of silence between the two users and generate prompts when the silence exceeded a certain threshold. This mechanism allowed the system to proactively introduce new topics before a conversation completely stalled. As a result, users were less likely to perceive awkward pauses and more likely to feel that the conversation flowed continuously. This anticipatory prompting likely contributed to the users' perception that conversations were easier to sustain under facilitated conditions.

Discussion on Q4 (Memory Recall)

A significant difference was found between Groups 1 and 2 in Q4, indicating that presenting questions based on reminiscence therapy was more effective in promoting memory recall. To further explore the underlying reasons for this effect, we analyzed participants' responses to Q8, which asked about the kinds of memories they recalled during the conversation.

Participants in Group 1 reported recalling a range of personally meaningful past experiences. For example, one participant described "memories of playing in the park until evening and childhood play," while another "remembered running to school every morning and the scenes of the high school grounds." Others reflected on memories from their junior and senior high school years, such as "school trips and high school club activities." These comments suggest that the system's prompts successfully encouraged conversation that naturally led to deeper reminiscence. Since local companions were physically walking through the university campus and therapy recipients were viewing this environment via video, the topic of school emerged organically. When such school-related dialogue continued, the system intervened to further steer the conversation toward recollections of student life, thereby facilitating even more extensive memory recall. Even when the initial

topic was not school-related, the system could introduce school-themed questions using contextual cues from the robot's camera, helping participants retrieve associated experiences.

In contrast, participants in Group 2 primarily discussed more recent or mundane topics. For example, one participant reported that they "talked about the most difficult part of living alone," while another recalled "some recent places I visited" but did not recall many past memories. Other responses included "I hardly recalled any past memories," and "I didn't talk about the past, so I didn't recall much." These findings suggest that conversations in Group 2 tended to revolve around current university life. Because the system's prompts were not designed to evoke older memories, participants' perception of memory recall was weaker compared to Group 1.

Taken together, these results indicate that even when users share the same environment, the use of reminiscence-based prompts can dramatically influence the content of the conversation. Comparing Groups 1 and 2, reminiscence-based prompts were more effective in guiding conversations toward older, contextually relevant, and emotionally significant memories. This suggests that, in addition to visual cues provided by video footage, prompting users to discuss meaningful topics from their past enhances the likelihood of reactivating those memories. As a result, the use of reminiscence prompts proved significantly more successful in facilitating memory recall.

Discussion on Q5 (Willingness to Reuse) and Number of Conversational Turns

The scores of Q5 were overall high in both conditions, and no significant difference was found between Group 1 and 2. This indicates that RIMER's core function—namely, situated remote conversation with facilitation of the robot that prevents conversations from stalling—was highly valued by participants.

Regarding the number of conversational turns, both facilitated groups (Groups 1 and 2) significantly outperformed the non-facilitated group (Group 3). Along the result with perceived ease of conversation (Q3), this provides strong evidence for the effectiveness of RIMER's core function; it successfully increased the quantity of dialogue and made it easier for participants to maintain conversational flow. Notably, there was no significant difference in these quantitative metrics between Group 1 and Group 2. This suggests that while any form of system intervention can sustain a conversation, the content of that intervention plays a more critical role in shaping its quality—as reflected in the results for conversational engagement (Q1) and memory recall (Q4).

5 Conclusion

In this paper, we proposed RIMER (Remote Interactive Memory Enhancement Robot), a shoulder-mounted facilitation robot system for remote reminiscence therapy that leverages large language models and reminiscence-based prompts. RIMER generates contextually appropriate questions for users by utilizing both the ongoing conversation between remote and local participants and real-time visual information captured by a camera mounted on the local companion's robot. The system is designed to increase the amount of speech and to facilitate memory recall for the therapy recipient.

Our findings indicate that reminiscence-based facilitation with RIMER was effective in both activating conversations and enhancing memory recall among therapy recipients. This effect can be

attributed to the fact that conversations centered around deeply meaningful and shared past experiences naturally led to more active and engaging interactions.

Looking ahead, we aim to further advance the system by developing methods for topic selection and question generation tailored to older adults' interests and shared experiences between users, as well as by integrating nonverbal cues for more adaptive support—all with the goal of contributing to dementia prevention.

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